

OPERATING SYSTEMS

UCCD 2103

FEB 2024

ASSIGNMENT REPORT

Part B

Group Number:

No.	Student ID	Name	Programme (CN/CS/CT/IA/IR)	E-mail
1				
2				
3				
4				

Assignment Part B (Linux Administration) 100 marks

Grouping: Please follow the same group members as you have formed in Part A

Assessment type: Individual (meaning each group member's answer will be evaluated individually)

Objective: To expose students to Linux OS administration, commands, bash scripting.

Due date/time: 5th May 2025 11.59 PM (link will be disabled automatically at 12.05 AM, 6th May 2025)

Submission:

Reports from all members should be combined into a single report and submit to WBLE. The link for Part B submission will be made available later. Only the group leader needs to submit one softcopy report, name the report as ***yourleadername.docx***. Your submission is final (no resubmission is allowed). Thus, please make sure you submit the correct file. Use the report cover page provided. Assignment free riders should be reported and excluded from the report submission.

Report formatting:

Your assignment report must use the cover page given in this document. For the content, use font size 12.

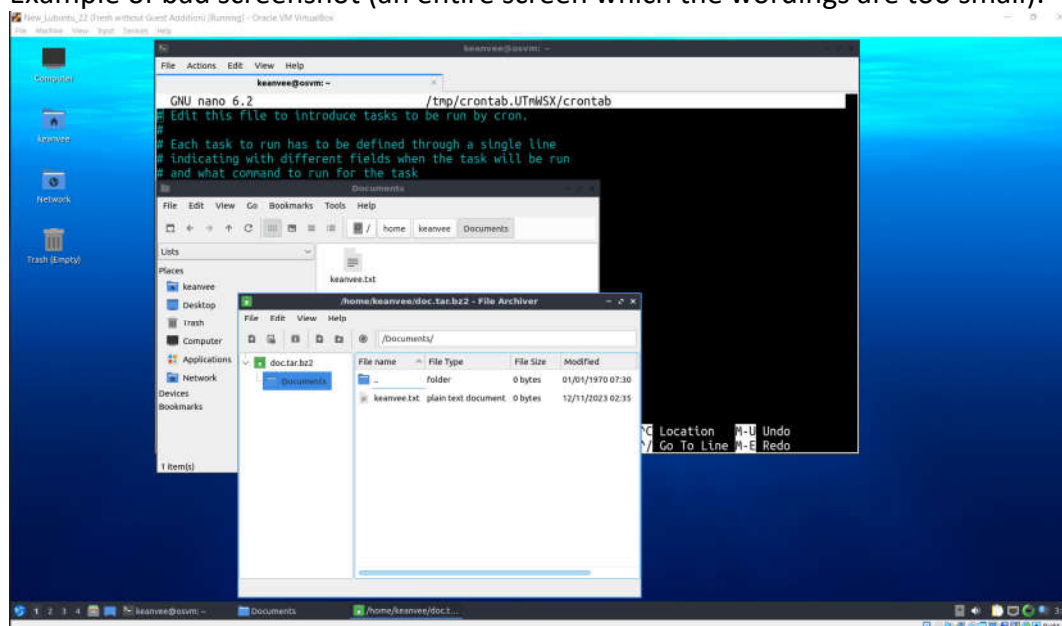
Reminder:

All the screenshots (except the VM settings) in the report must contain / able to show your user name to **proof the originality of your work**. Notice that the screenshots included in this document show my user name "keanvee".

Any group reports that have the **same screenshots or usernames**, will be counted as **plagiarism** and be awarded 0 marks.

The content of all screenshots must be clear enough to be read. DO NOT take the screenshot of the entire screen.

Example of bad screenshot (an entire screen which the wordings are too small):



Your solution files:

Inside the zip file that you submit to WBLE should contain these files:

- One report in PDF format:
 - Cover page:
 - Answer for group member 1 (state the name and ID):
 - Content of Task 1
 - Content of Task 2
 - Answer for group member 2 (state the name and ID):
 - Content of Task 1
 - Content of Task 2
 - Answer for group member 3 (state the name and ID):
 - Content of Task 1
 - Content of Task 2
 - Answer for group member 4 (state the name and ID):
 - Content of Task 1
 - Content of Task 2
- Group member 1 folder:
 - CPU.sh (the shell script file)
 - CPU-stop.sh (the shell script file)
 - CPU.csv (the log file)
- Group member 2 folder:
 - RAM.sh (the shell script file)
 - RAM-stop.sh (the shell script file)
 - RAM.csv (the log file)
- Group member 3 folder:
 - VM.sh (the shell script file)
 - VM-stop.sh (the shell script file)
 - VM.csv (the log file)
- Group member 4 folder:
 - network.sh (the shell script file)
 - network-stop.sh (the shell script file)
 - network.csv (the log file)

Task 1: Installing Linux OS (30 marks)

Each group member is required to **install the Linux OS** in a virtual machine. In fact, if you have started answering Part A, you should have done this step, and you can use the same VM you use in Part A as the answer for this task. If you have not, then you can use VirtualBox or VMware Workstation Player and install Ubuntu OS. Create a virtual machine with the following specification:

Name of virtual machine: ***your_name_vm***

Processor: **2**

Memory: **2GB** (can be less or more based the available memory in the host machine)

HD storage: **20 GB**

Recommended OS (for consistency of guides in this assignment): **Ubuntu 22.04**

User name: ***Your_name***

Host (machine) name: **osvm**

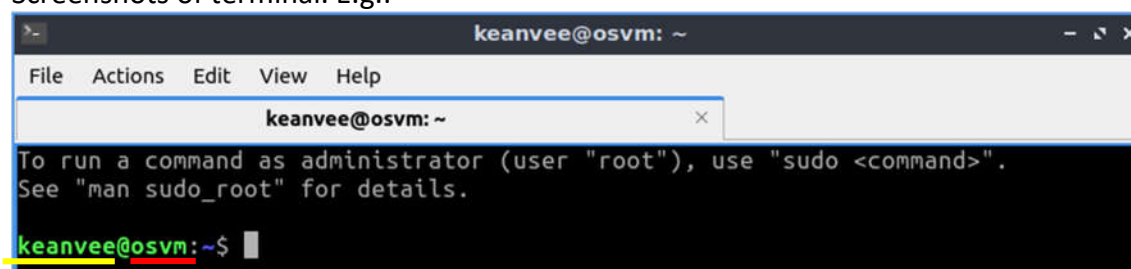
NOTE: Using Kali Linux virtual machine image provided to you in another course does not count as “installation” and is prohibited in this assignment.

Please take note that when the VM boots from the ISO file, the OS is running in Live mode, which means the OS only resides in RAM for temporary usage, with no permanent storage. All the data or changes will be gone once the VM is shut down or reboot. Thus, you need to make sure you have done the installation. Otherwise, you will not be awarded the installation marks.

To be included in the report:

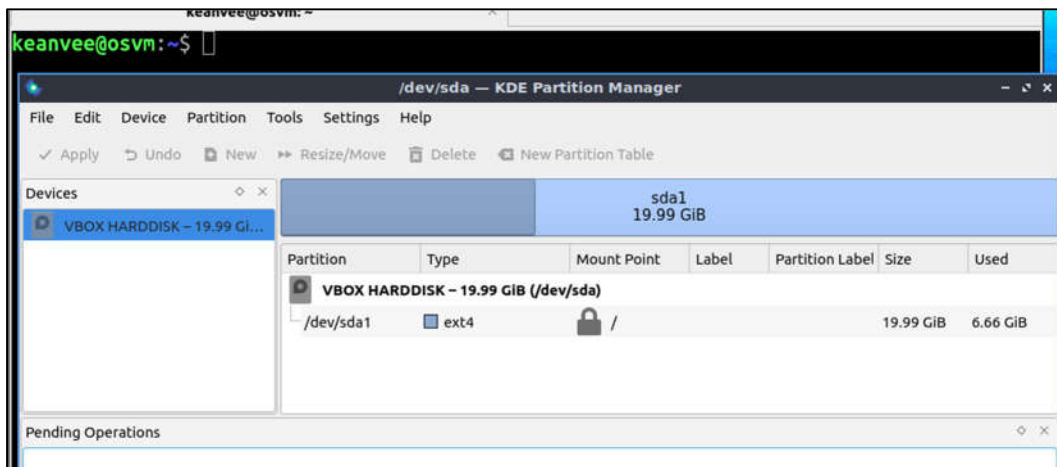
The following screenshots, from each member, should be included in the report.

1. Screenshots of terminal. E.g.:



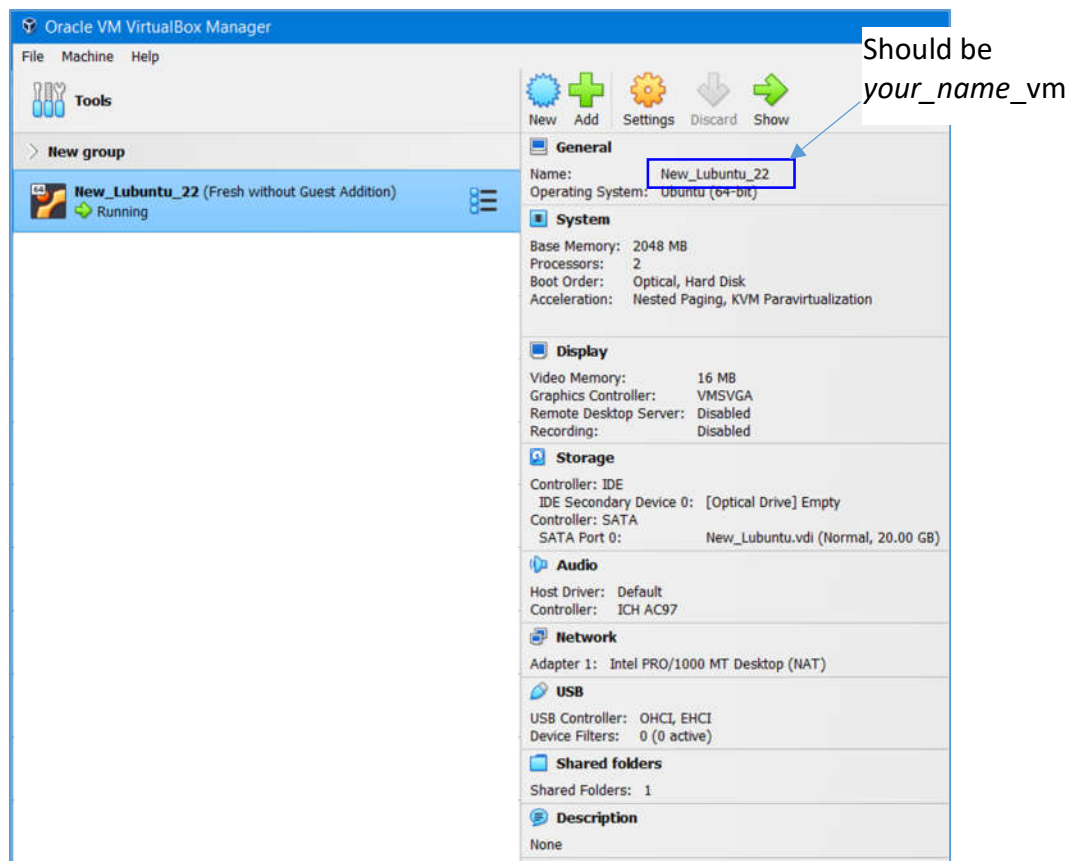
User name host name

2. The partition information. E.g. (Ubuntu):
Notice that the user name is shown in the terminal behind the partition manager.

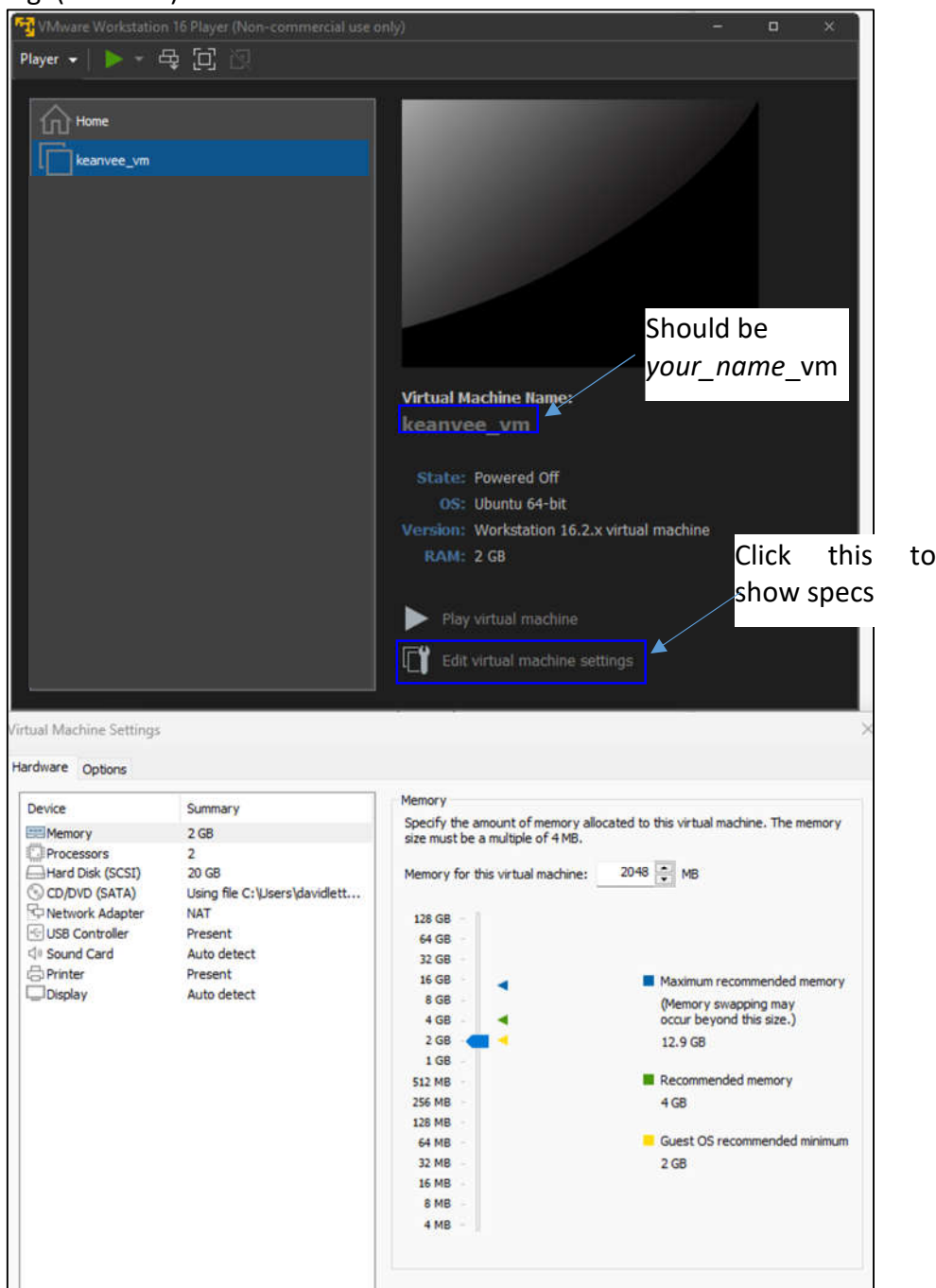


Note: It may look different for other distros.

3. The specs of the Virtual Machine. E.g. (VirtualBox):



E.g. (VMware):



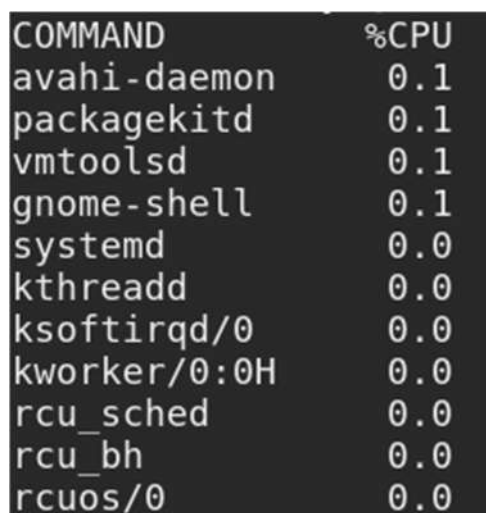
Task 2: CPU resource monitoring using Bash script (70 marks)

Before attempting Task 2 and beyond, go through the `bash` tutorial in the PDF file. It has been uploaded to WBLE. Basically, a shell script is consisting of lines of `bash` commands (those commands you typed in Ubuntu terminal are `bash` commands). These commands can be combined into a script so that task automation can be realized when you have a task that involves a string of commands. You do not want to manually input these commands into the terminal line by line due to inconvenience.

Then, choose one of the following questions. Each group member must choose a different question from one another (group members cannot attempt the same question). Every question has the same difficulty, and you can still discuss among yourselves even though they are different questions. Their differences are only the measured metrics and command used.

1. CPU resource monitoring program

Design a CPU resource monitoring program using a shell script. The objective of your shell script is to capture the task name with the highest CPU resource utilization, in terms of percentage and record it into a file called `CPU.csv`. You must use `ps` command to accomplish this question. Get familiar with how `ps` command is used. Hint: you can execute `ps` and use `-eo` option to make it outputs the information that you want. In this question, the output we are interested are task names (the column called command) and %CPU, as shown in Figure 1a.



COMMAND	%CPU
avahi-daemon	0.1
packagekitd	0.1
vmtoolsd	0.1
gnome-shell	0.1
systemd	0.0
kthreadd	0.0
ksoftirqd/0	0.0
kworker/0:0H	0.0
rcu_sched	0.0
rcu_bh	0.0
rcuos/0	0.0

Figure 1a: An excerpt of an output for the command `ps` sorted according to %CPU in descending order

Use the following guides to write a shell script called `CPU.sh`:

- Before begin monitoring, the shell script shall clear the previous content in `CPU.csv` file. (5 marks)
- Use `date` command (with the suitable command option) to get the current time. The format of the output should be `hr:min:sec` (refer to the screenshot in Figure 1b) (10 marks)
- The `ps` command displays a lot of information. But you are interested to capture the name of the task that has the **highest** percentage of CPU utilization (%CPU). To do this, please sort the output of the `ps` command so that the %CPU is in descending order. For example, in Figure 1a, task `avahi-daemon` has the highest %CPU utilization, and that task name shall be recorded. (10 marks)
- The script shall also record down the percentage of CPU utilization (%CPU) for that task you filtered in step 1. For example, in Figure 1a, the value `0.1` (%CPU of `avahi-daemon`) shall be recorded. (10 marks)
- For a single reading, store the values you filtered in steps (b) to (d), into one row, on a file called `CPU.csv`. Separate each value by a semicolon (;). This is called a delimiter. The next reading shall continue at the next line, as shown in the example in Figure 1b. (5 marks)
- Your script will run non-stop (i.e. using an infinite loop) and can only be stopped by the user using Ctrl-c keys. (5 marks)
- Your script must be able to let the user to determine the period (how frequent) of monitoring the CPU reading by inserting the period (in second) as the first

input parameter. For example, if the user intends to record down the reading every 2 seconds, the command shall be:

```
./CPU.sh 2
```

If the user intends to record down the reading every 4 seconds, the command shall be: `./CPU.sh 4`

If the user does not enter the first input parameter, your script shall make the period is set to a default value of 1 minute (60 seconds). (10 marks)

For steps (a) to (g), show the content of `CPU.sh` and `CPU.csv` in your report.

To test, it is working, you may use your Firefox browser to play a Youtube video so that the CPU utilization is > 0%.

A sample `CPU.csv` is given below, showing the first four lines of the file for command: `./CPU.sh 4`

i.e 0.1% is the CPU utilization of task `avahi-daemon` at time 23:21:50 and the shell script will measure again after 4 seconds etc. Hint: Sometimes the duration between two readings is not accurate and may lag by 1 second. This is normal.

```
23:21:50 ; avahi-daemon; 0.1
23:21:54 ; avahi-daemon; 0.1
23:21:58 ; avahi-daemon; 0.1
23:22:02 ; avahi-daemon; 0.1
23:22:06 ; avahi-daemon; 0.1
23:22:10 ; avahi-daemon; 0.1
```

Figure 1b: A sample content of readings recorded in `CPU.csv` where in each row, the first, second, and third columns are time of the reading being recorded, the task name with the highest %CPU consumption, and its percentage of CPU consumption, respectively.

- h) You are required to make your resource monitoring script to run in the background of terminal. To do so, put the `&` character at the end of the shell script command: `./CPU.sh 4 &`

After that you may use `ps -all` command to check if `CPU.sh` is running.

(5 marks)

- i) Finally, you need to figure out how to stop your background script. Design a shell script, called `CPU-stop.sh` to stop it. Verify your solution is working.

Hint: You can explore how to use `kill` command.

(10 marks)

For step (h), show the screenshot in your report. For step (i), show the content of `CPU-stop.sh` and the verification screenshot in your report.

2. RAM resource monitoring program

Design a RAM resource monitoring program using a shell script. The objective of your shell script is to capture the task name with the highest RAM utilization, in terms of percentage and record it into a file called `RAM.csv`. You must use `ps` command to accomplish this question. Get familiar with how `ps` command is used. Hint: you can execute `ps` and use `-eo` option to make it outputs the information that you want. In this question, the output we are interested are task names (the column called `COMMAND`) and `%MEM`, as shown in Figure 2a.



COMMAND	%MEM
packagekitd	9.4
gedit	4.0
gnome-shell	3.9
gnome-shell	2.8
gnome-software	2.8
Xorg	1.5
gnome-settings-	1.3
nautilus	1.2
vmtoolsd	1.2
gnome-settings-	1.1
fwupd	1.0
evolution-alarm	1.0
evolution-calen	1.0
evolution-calen	0.9

Figure 2a: An excerpt of an output for the command `ps` sorted according to `%MEM` in descending order

Use the following guides to write a shell script called `RAM.sh`:

- Before begin monitoring, the shell script shall clear the previous content in `RAM.csv` file. (5 marks)
- Use `date` command (with the suitable command option) to get the current time. The format of the output should be `hr:min:sec` (refer to the screenshot in Figure 2b) (10 marks)
- The `ps` command displays a lot of information. But you are interested to capture the name of the task that has the **highest** percentage of memory utilization (`%MEM`). To do this, please sort the output of the `ps` command so that the `%MEM` is in descending order. For example, in Figure 2a, task `packagekitd` has the highest `%MEM` utilization, and that task name shall be recorded. (10 marks)
- The script shall also record down the percentage of MEM utilization (`%MEM`) for that task you filtered in step 1. For example, in Figure 2a, the value `9.4` (`%MEM` of `packagekitd`) shall be recorded. (10 marks)
- For a single reading, store the values you filtered in steps (b) to (d), into one row, on a file called `RAM.csv`. Separate each value by a semicolon (;). This is called a delimiter. The next reading shall continue at the next line, as shown in the example in Figure 2b. (5 marks)

- f) Your script will run non-stop (i.e. using an infinite loop) and can only be stopped by the user using Ctrl-c keys. (5 marks)
- g) Your script must be able to let the user to determine the period (how frequent) of monitoring the `MEM` reading by inserting the period (in second) as the first input parameter. For example, if the user intends to record down the reading every 2 seconds, the command shall be:
`./RAM.sh 2`
 If the user intends to record down the reading every 4 seconds, the command shall be: `./RAM.sh 4`
 If the user does not enter the first input parameter, your script shall make the period is set to a default value of 1 minute (60 seconds). (10 marks)

For steps (a) to (g), show the content of `RAM.sh` and `RAM.csv` in your report.

To test it is working, you may use your Firefox browser and open a few tabs of websites. You may play a Youtube video in one of the tabs so that there are some changes in your RAM utilization.

A sample `RAM.csv` is given below, showing the first four lines of the file for command: `./RAM.sh 4`
 i.e `packagekitd` utilizes 9.4% of system RAM at time 23:46:41 and the shell script will it measure again after 4 seconds.

```
23:46:41 ; packagekitd; 9.4
23:46:45 ; packagekitd; 9.4
23:46:49 ; packagekitd; 9.4
23:46:53 ; packagekitd; 9.4
23:46:57 ; packagekitd; 9.4
```

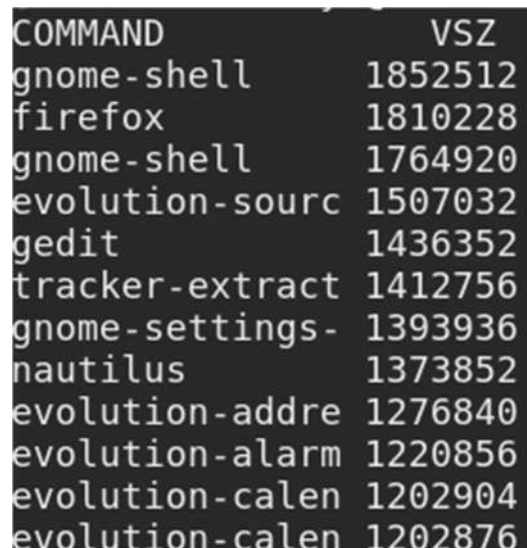
Figure 2b: A sample content of readings recorded in `RAM.csv` where in each row, the first, second, and third columns are the time of the reading being recorded, the task name with the highest `%MEM` consumption, and its percentage of memory consumption, respectively.

- h) You are required to make your resource monitoring script to run in the background of terminal. To do so, put the `&` character at the end of the shell script command: `./RAM.sh 4 &`
 After that you may use `ps -all` command to check if `RAM.sh` is running. (5 marks)
- i) Finally, you need to figure out how to stop your background script. Design a shell script, called `RAM-stop.sh` to stop it. Verify your solution is working.
 Hint: You can explore how to use `kill` command. (10 marks)

For step (h), show the screenshot in your report. For step (i), show the content of `RAM-stop.sh` and the verification screenshot in your report.

3. Virtual Memory resource monitoring program

Design a Virtual Memory resource monitoring program using a shell script. The objective of your shell script is to capture the task name with the **highest** virtual memory consumption, in bytes and record it into a file called `VM.csv`. You must use `ps` command to accomplish this question. Get familiar with how `ps` command is used. Hint: you can execute `ps` and use `-eo` option to make it outputs the information that you want. In this question, the output we are interested are task names (the column called `COMMAND`) and `VSZ` (VIRTUAL MEMORY SIZE) as shown in Figure 3a.



COMMAND	VSZ
gnome-shell	1852512
firefox	1810228
gnome-shell	1764920
evolution-sourc	1507032
gedit	1436352
tracker-extract	1412756
gnome-settings-	1393936
nautilus	1373852
evolution-addre	1276840
evolution-alarm	1220856
evolution-calen	1202904
evolution-calen	1202876

Figure 3a: An excerpt of an output for the command `ps` sorted according to `VSZ` in descending order

Use the following guides to write a shell script called `VM.sh`:

- Before begin monitoring, the shell script shall clear the previous content in `VM.csv` file. (5 marks)
- Use `date` command (with the suitable command option) to get the current time. The format of the output should be `hr:min:sec` (refer to the screenshot in Figure 3b) (10 marks)
- The `ps` command displays a lot of information. But you are interested to capture the name of the task that has the **highest** virtual memory consumption (`VSZ`). To do this, please sort the output of the `ps` command so that the `VSZ` is in descending order. For example, in Figure 3a, task `gnome-shell` has the highest `VSZ` utilization, and that task name shall be recorded. (10 marks)
- The script shall also record down the virtual memory utilization (`VSZ`) for that task you filtered in step b. For example, in Figure 3a, the value 1810228 (`VSZ` of `gnome-shell`) shall be recorded. (10 marks)
- For a single reading, store the values you filtered in steps (b) to (d), into one row, on a file called `VM.csv`. Separate each value by a semicolon (;). This is called a delimiter. The next reading shall continue at the next line, as shown in the example in Figure 3b. (5 marks)
- Your script will run non-stop (i.e. using an infinite loop) and can only be stopped by the user using `Ctrl-c` keys. (5 marks)

- g) Your script must be able to let the user to determine the period (how frequent) of monitoring the VSZ reading by inserting the period (in second) as the first input parameter. For example, if the user intends to record down the reading every 2 seconds, the command shall be:

```
./VM.sh 2
```

If the user intends to record down the reading every 4 seconds, the command shall be: `./VM.sh 4`

If the user does not enter the first input parameter, your script shall make the period is set to a default value of 1 minute (60 seconds). (10 marks)

For steps (a) to (g), show the content of `VM.sh` and `VM.csv` in your report.

To test it is working, you may use your Firefox browser and open a few tabs of websites. You may play a Youtube video in one of the tabs so that there are some changes in your virtual memory utilization.

A sample `VM.csv` is given below, showing the first four lines of the file for command: `./VM.sh 4`

i.e `gnome-shell` utilizes 1852580 Bytes of Virtual Memory at time 19:19:52 and the shell script will it measure again after 4 seconds.

```
19:19:52 ; gnome-shell; 1852580
19:19:56 ; gnome-shell; 1852580
19:20:00 ; gnome-shell; 1852580
19:20:04 ; gnome-shell; 1852580
19:20:08 ; gnome-shell; 1852580
19:20:12 ; firefox; 1870996
19:20:16 ; firefox; 1874932
```

Figure 3b: A sample content of readings recorded in `VM.csv` where in each row, the first, second, and third columns are the time of the reading being recorded, the task name with the highest VSZ consumption, and its VSZ consumption in bytes, respectively.

- h) You are required to make your resource monitoring script to run in the background of terminal. To do so, put the `&` character at the end of the shell script command: `./VM.sh 4 &`

After that you may use `ps -all` command to check if `VM.sh` is running.

(5 marks)

- j) Finally, you need to figure out how to stop your background script. Design a shell script, called `VM-stop.sh` to stop it. Verify your solution is working.

Hint: You can explore how to use `kill` command.

(10 marks)

For step (h), show the screenshot in your report. For step (i), show the content of `VM-stop.sh` and the verification screenshot in your report.

4. Network usage monitoring program

Design a network usage monitoring program using a shell script. The objective of your shell script is to capture total network consumption, in kilobytes (KB) and record it into a file called `network.csv`. You must use `ip` command to accomplish this question. Get familiar with how `ip` command is used. Hint: you can execute `ip` on a specific network interface (`ens33` is normally the default interface name if you run your Linux on the VMware) to capture the total received (RX) data and transmitted (TX) data, respectively. In this question, the output we are interested are RX and TX in bytes, as shown in Figure 4a.

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 00:0c:29:5a:e3:f0 brd ff:ff:ff:ff:ff:ff
    RX: bytes  packets errors dropped missed  mcast
       1762711679 1208521      0      0      0      0
    TX: bytes  packets errors dropped carrier collsns
       10683838  140130      0      0      0      0
    altnam enp2s1
```

Figure 4a: An excerpt of an output for the command `ip` on `ens33` network interface.

In this example, the RX and TX readings at that point are 1762711679 bytes and 10683838 bytes, respectively.

Use the following guides to write a shell script called `network.sh`:

- Before begin monitoring, the shell script shall clear the previous content in `network.csv` file. (5 marks)
- Use `date` command (with the suitable command option) to get the current time. The format of the output should be `hr:min:sec` (refer to the screenshot in Figure 4b) (10 marks)
- The `ip` command displays a lot of information. But you are interested to capture the total received (RX) and transmitted data (TX) size, which are in bytes. To do this, you need to capture these two values. (10 marks)
- The script shall also convert both values into kilobytes. (10 marks)
- For a single reading, store the RX and TX values you filtered in steps (b) to (d), into one row, on a file called `network.csv`. Separate each value by a semicolon (;). This is called a delimiter. The next reading shall continue at the next line, as shown in the example in Figure 4b. (5 marks)
- Your script will run non-stop (i.e. using an infinite loop) and can only be stopped by the user using Ctrl-c keys. (5 marks)
- Your script must be able to let the user to determine the period (how frequent) of monitoring the RX and TX readings by inserting the period (in second) as the first input parameter. For example, if the user intends to record down the readings every 2 seconds, the command shall be:

```
./network.sh 2
```

If the user intends to record down the readings every 4 seconds, the command shall be: `./network.sh 4`

If the user does not enter the first input parameter, your script shall make the period is set to a default value of 1 minute (60 seconds). (10 marks)

For steps (a) to (g), show the content of `network.sh` and `network.csv` in your report.

To test it is working, you may use your Firefox browser and open a few tabs of websites. You may play a Youtube video in one of the tabs so that there are some changes in your network utilization.

A sample `network.csv` is given below, showing the first four lines of the file for command: `./network.sh 4`

i.e the total network utilization so far at time 14:19:08 is 1721412 KB (RX) and 10444 KB (TX) and the shell script will measure it again after 4 seconds.

```
14:59:08; 1721412; 10444
14:59:12; 1721412; 10444
14:59:16; 1721412; 10444
14:59:20; 1721838; 10507
14:59:24; 1721861; 10534
14:59:28; 1722331; 10559
14:59:32; 1722334; 10561
14:59:36; 1722335; 10561
```

Figure 4b: A sample content of readings recorded in `network.csv` where in each row, the first, second, and third columns are the time of the reading being recorded, the RX value, and the TX consumption in kilobytes, respectively.

- h) You are required to make your resource monitoring script to run in the background of terminal. To do so, put the `&` character at the end of the shell script command: `./network.sh 4 &`
After that you may use `ps -all` command to check if `network.sh` is running. (5 marks)
- i) Finally, you need to figure out how to stop your background script. Design a shell script, called `network-stop.sh` to stop it. Verify your solution is working.
Hint: You can explore how to use `kill` command. (10 marks)

For step (h), show the screenshot in your report. For step (i), show the content of `network-stop.sh` and the verification screenshot in your report.